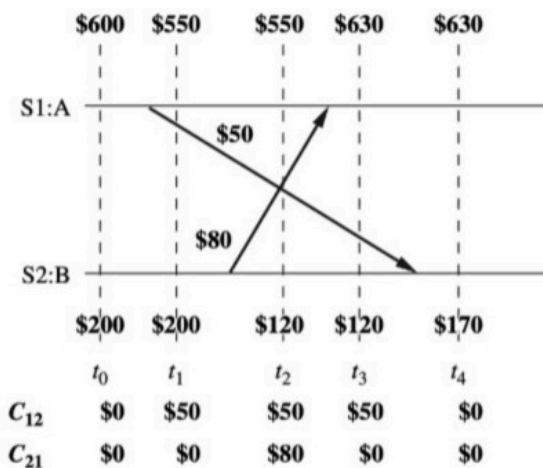


Distributed Systems

Slip Test 1

Question 1 (3 Marks)

Slip Test – Question 1



Consider the transaction shown.
Process A initiates a snapshot at time t_0 .
Assume the marker reaches process B
at time t_2 .

Provide details of the global snapshot
collected by Process A

Global Snapshot Details: The Initiator (Process A) will collect the following global state:

1. Local State of Process A: \$600

- Reasoning: A initiates the snapshot at t_0 , before any transactions occur.
2. Local State of Process B: \$120
 - Reasoning: The marker reaches B at t_2 . At this timestamp B has already sent the \$80, leaving it with \$120.
 3. State of Channel A $\rightarrow$ B: Empty (\$0)
 - Reasoning: The \$50 message from A was sent at t_1 (after A's snapshot at t_0). Therefore, it is not part of the snapshot for this channel.
 4. State of Channel B $\rightarrow$ A: \$80
 - Reasoning: B sent the \$80 message before receiving marker at t_2 . However, A receives this message at t_3 but A's snapshot has already being taken at t_0 . Therefore, it must be recorded in the channel state.

Marking Scheme:

- 1 mark for Local State of A
- 1 mark for Local State of B
- 1 mark for State of Channel B $\rightarrow$ A

Question 2 (4.5 marks)

A) Can Chandy Lamport snapshot algorithm used for FIFO channel also be used for causal messaging framework? (1.5 marks)

- Answer: True

Reasoning: Causal Delivery is a stricter consistency model than FIFO (i.e., Causal Delivery $\implies$ FIFO). Any system that guarantees Causal ordering inherently guarantees FIFO ordering for point-to-point links. Therefore, the Chandy-Lamport algorithm will function correctly and consistently on a causal messaging framework without modification.

Marking Scheme:

- Full marks only with correct explanation.

- 0 marks if written just "True/False"

B) Can Acharya Badrinath snapshot algorithm for Causal messaging also be used for FIFO channels? (3 marks)

- Answer: False

Reasoning: It relies on this property to ensure that if a message is delivered, all causally preceding messages have also been delivered. If you run this algorithm on a system that only provides FIFO channels, the algorithm will fail. Consider below:

- Process P_1 sends message M_1 to P_2 , and then sends a message to P_3 which triggers P_3 to send M_2 to P_2 . Causally, $M_1 \rightarrow M_2$.
- In a simple FIFO system, P_2 guarantees it receives messages **from the same sender** in order. However, it does not guarantee order across different senders. P_2 could receive M_2 (from P_3) *before* M_1 (from P_1). This scenario could lead to inconsistent snapshot.

Marking Scheme:

- Full marks only with counter example.
- Explanation without counter example receives 1.5 marks.
- 0 marks if written just "True/False".